In [9]:
```
pip install pandas
```

```
Requirement already satisfied: pandas in c:\users\acer\anaconda3\lib\site-packages (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\acer\anaconda3\lib\site-packages (from pandas) (2.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\acer\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\acer\anaconda3\lib\site-packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\acer\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\acer\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [10]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [12]:
```
df = pd.read_excel(r'C:\Users\ACER\Downloads\shark_tank.xlsx')
```

In [13]:
```
print(df.head())
```

```
   Season Number Season Start Season End  Episode Number  \
0              1   2021-12-20 2022-02-04               1
1              1   2021-12-20 2022-02-04               1
2              1   2021-12-20 2022-02-04               1
3              1   2021-12-20 2022-02-04               2
4              1   2021-12-20 2022-02-04               2

                Episode Title  Pitch Number         Startup Name  \
0  Badlegi Business Ki Tasveer             1      Blue Pine Foods
1  Badlegi Business Ki Tasveer             2        Booz Scooters
2  Badlegi Business Ki Tasveer             3  Heart Up My Sleeves
3      Insaan, Ideas Aur Sapne             4           Tagz Foods
4      Insaan, Ideas Aur Sapne             5       Head And Heart

                      Industry  \
0                         Food
1  Vehicles/Electrical Vehicles
```

```
2                 Beauty/Fashion
3                           Food
4                      Education

                            Business Description  \
0                                   Frozen Momos
1  Renting E-Bike For Mobility In Private Spaces
2                             Detachable Sleeves
3                     Healthy Potato Chips Snacks
4                        Brain Development Course

                 Company Website  ...  Ghazal Investment Amount  \
0     https://bluepinefoods.com/  ...                       NaN
1        https://www.boozup.net/  ...                       NaN
2  https://heartupmysleeves.com/  ...                       NaN
3         https://tagzfoods.com/  ...                       NaN
4             https://thehnh.in/  ...                       NaN

   Ghazal Investment Equity  Ghazal Debt Amount  Amit Investment Amount  \
0                       NaN                 NaN                     NaN
1                       NaN                 NaN                     NaN
2                       NaN                 NaN                     NaN
3                       NaN                 NaN                     NaN
4                       NaN                 NaN                     NaN

   Amit Investment Equity Amit Debt Amount Guest Investment Amount  \
0                     NaN              NaN                     NaN
1                     NaN              NaN                     NaN
2                     NaN              NaN                     NaN
3                     NaN              NaN                     NaN
4                     NaN              NaN                     NaN

  Guest Investment Equity Guest Debt Amount  Guest Name
0                     NaN               NaN         NaN
1                     NaN               NaN         NaN
2                     NaN               NaN         NaN
3                     NaN               NaN         NaN
4                     NaN               NaN         NaN
```

```
[5 rows x 64 columns]
```

In [14]:

```python
print(df.describe())
```

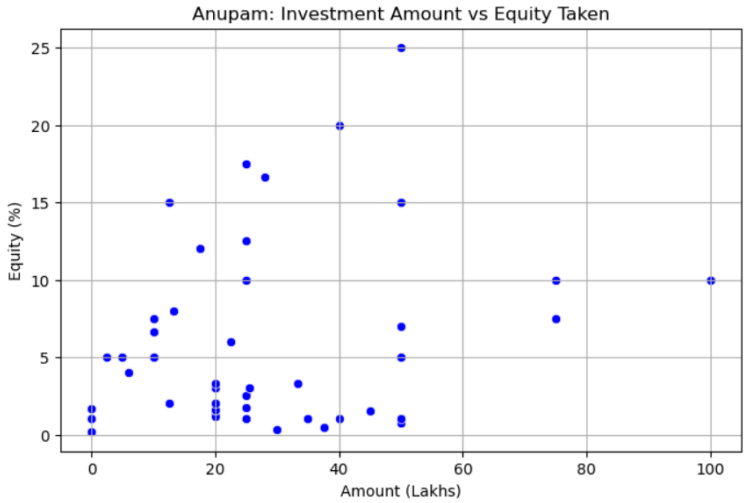
```
       Season Number                   Season Start  \
count     290.000000                            290
mean        1.579310  2022-07-26 23:30:12.413793024
min         1.000000            2021-12-20 00:00:00
25%         1.000000            2021-12-20 00:00:00
50%         2.000000            2023-01-02 00:00:00
75%         2.000000            2023-01-02 00:00:00
max         2.000000            2023-01-02 00:00:00
std         0.494523                            NaN

                         Season End  Episode Number  Pitch Number  \
count                           290      290.000000    290.000000
mean   2022-09-23 03:28:33.103448320       23.779310    162.879310
min             2022-02-04 00:00:00        1.000000      1.000000
25%             2022-02-04 00:00:00       13.000000     73.250000
50%             2023-03-10 00:00:00       23.000000    175.500000
75%             2023-03-10 00:00:00       34.000000    247.750000
max             2023-03-10 00:00:00       51.000000    320.000000
std                             NaN       13.585291     96.851486

       Number of Presenters  Male Presenters  Female Presenters  \
count            290.000000       290.000000         290.000000
mean               2.086207         1.524138           0.558621
min                1.000000         0.000000           0.000000
25%                2.000000         1.000000           0.000000
50%                2.000000         1.000000           0.000000
75%                2.750000         2.000000           1.000000
max                6.000000         6.000000           3.000000
std                0.858188         1.012604           0.679669
```

```
       Transgender Presenters  Couple Presenters  ...  Peyush Debt Amount  \
count              290.000000         287.000000  ...           11.000000
mean                 0.003448           0.181185  ...           31.090909
min                  0.000000           0.000000  ...           10.000000
25%                  0.000000           0.000000  ...           23.500000
50%                  0.000000           0.000000  ...           25.000000
75%                  0.000000           0.000000  ...           40.000000
max                  1.000000           1.000000  ...           60.000000
std                  0.058722           0.385844  ...           15.332675

       Ghazal Investment Amount  Ghazal Investment Equity  Ghazal Debt Amount  \
count                  7.000000                  7.000000                 0.0
mean                  18.570036                  6.671429                 NaN
min                    0.000253                  1.000000                 NaN
25%                   13.330000                  3.100000                 NaN
50%                   20.000000                  5.000000                 NaN
75%                   25.000000                  8.500000                 NaN
max                   33.330000                 17.500000                 NaN
std                   10.988788                  5.717434                 NaN

       Amit Investment Amount  Amit Investment Equity  Amit Debt Amount  \
count               21.000000               21.000000          4.000000
mean                38.138571                5.190314         36.250000
min                  5.000000                1.000000         10.000000
25%                 18.750000                1.500000         21.250000
50%                 30.000000                3.000000         35.000000
75%                 50.000000                5.000000         50.000000
max                100.000000               20.000000         65.000000
std                 26.305707                5.599211         23.935678
```

```
       Guest Investment Amount  Guest Investment Equity  Guest Debt Amount
count                 2.000000                 2.000000           2.000000
mean                 32.750000                 1.500000          23.500000
min                  25.000000                 0.500000          12.500000
25%                  28.875000                 1.000000          18.000000
50%                  32.750000                 1.500000          23.500000
75%                  36.625000                 2.000000          29.000000
max                  40.500000                 2.500000          34.500000
std                  10.960155                 1.414214          15.556349

[8 rows x 53 columns]
```

In [17]:

```python
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x='Anupam Investment Amount', y='Anupam Investment Equity', color='blue')

plt.title('Anupam: Investment Amount vs Equity Taken')
plt.xlabel('Amount (Lakhs)')
plt.ylabel('Equity (%)')
plt.grid(True)
plt.show()
```



In [25]:

```
# 1. Find the maximum value first
max_investment = df['Anupam Investment Amount'].max()

# 2. Filter the dataframe to find the row with that max value
max_invested_company = df[df['Anupam Investment Amount'] == max_investment]

# 3. Print the name
print("The company with the maximum investment from Anupam is:")
print(max_invested_company[['Startup Name', 'Anupam Investment Amount']])
```

```
The company with the maximum investment from Anupam is:
    Startup Name  Anupam Investment Amount
148  Zillionaire                     100.0
```

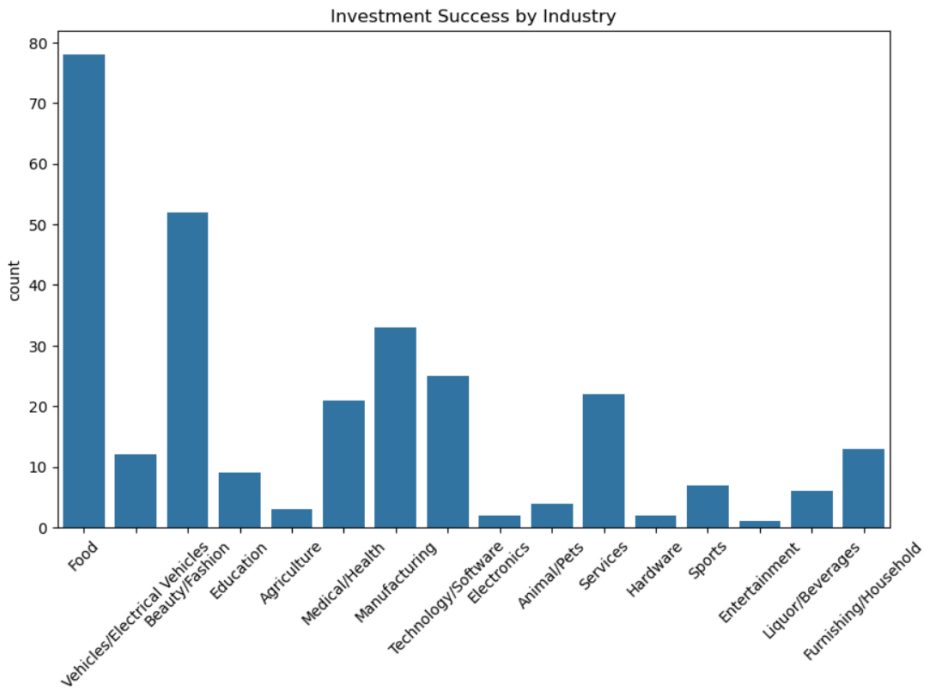
In [26]:

```
# 1. Find the highest equity percentage he took
max_equity = df['Anupam Investment Equity'].max()

# 2. Filter the dataframe to find the startup(s)
max_equity_startup = df[df['Anupam Investment Equity'] == max_equity]

# 3. Print the name and the equity percentage
print("Startup(s) where Anupam took the maximum equity:")
print(max_equity_startup[['Startup Name', 'Anupam Investment Equity']])
```

```
Startup(s) where Anupam took the maximum equity:
    Startup Name  Anupam Investment Equity
172        Raasa                      25.0
```

In [21]:

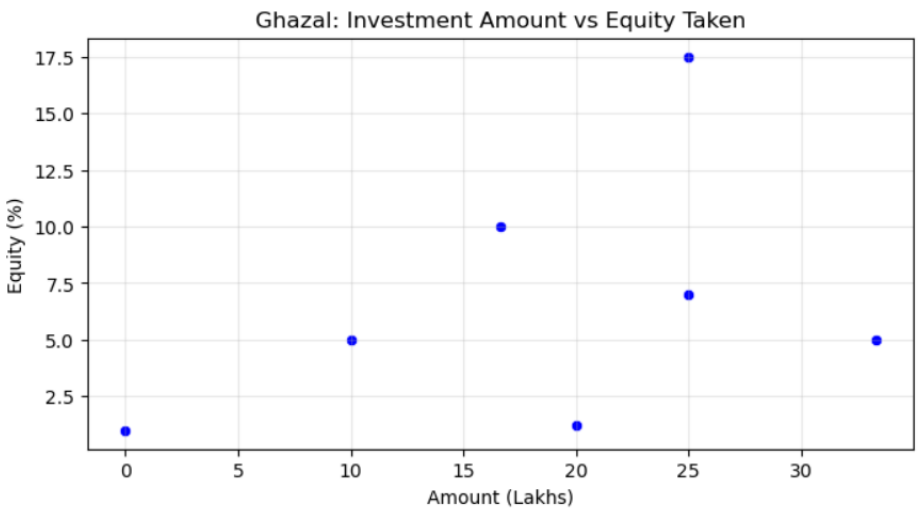
```
#bar chart showing the number of pitches per industry
plt.figure(figsize=(10,6))
sns.countplot(data=df, x='Industry')
plt.xticks(rotation=45)
plt.title('Investment Success by Industry')
plt.show()
```

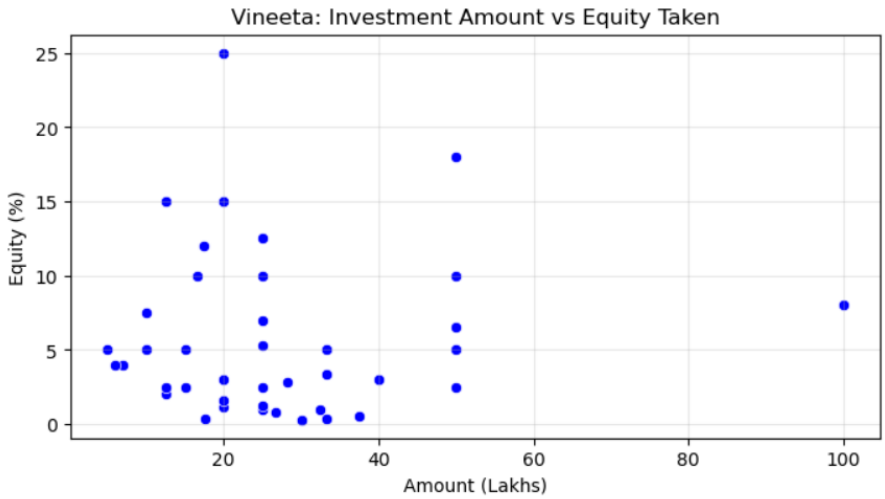


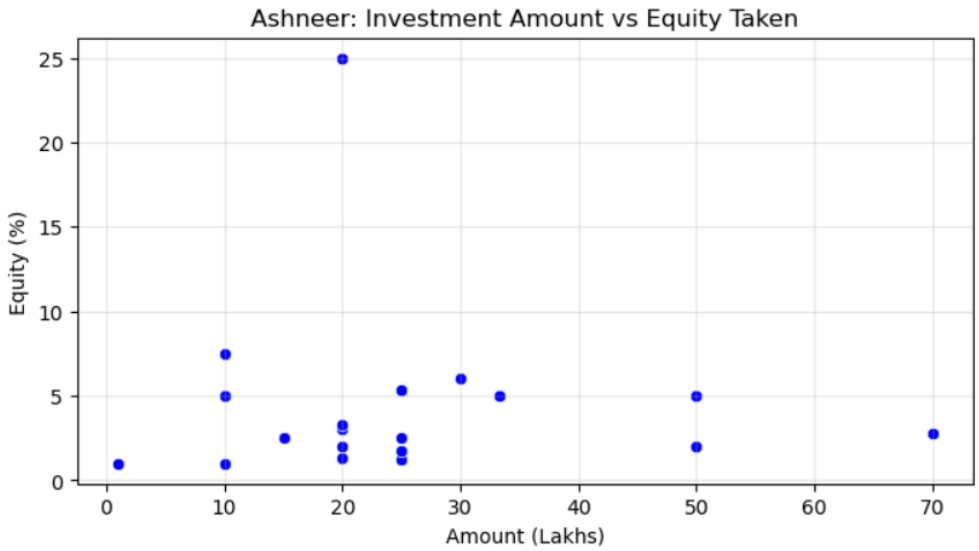
Industry

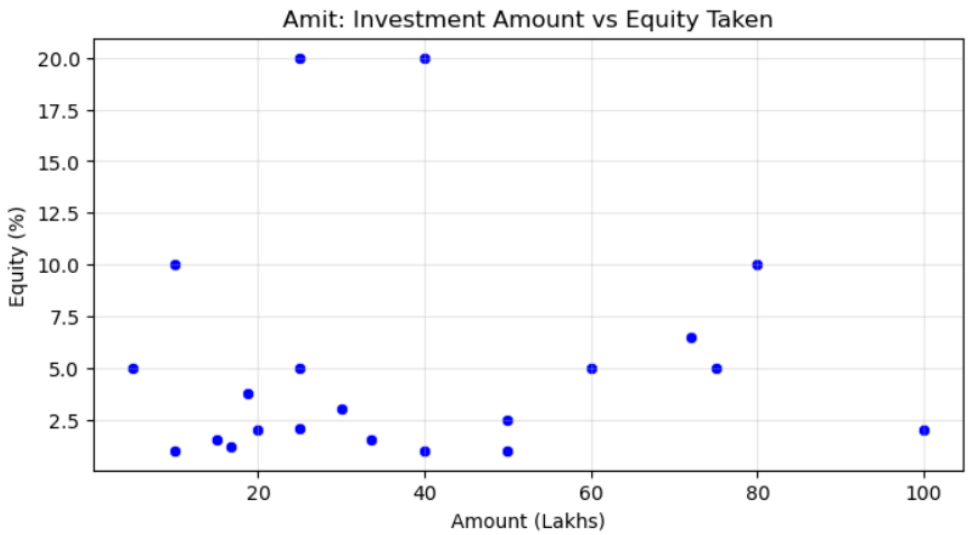
In [31]:

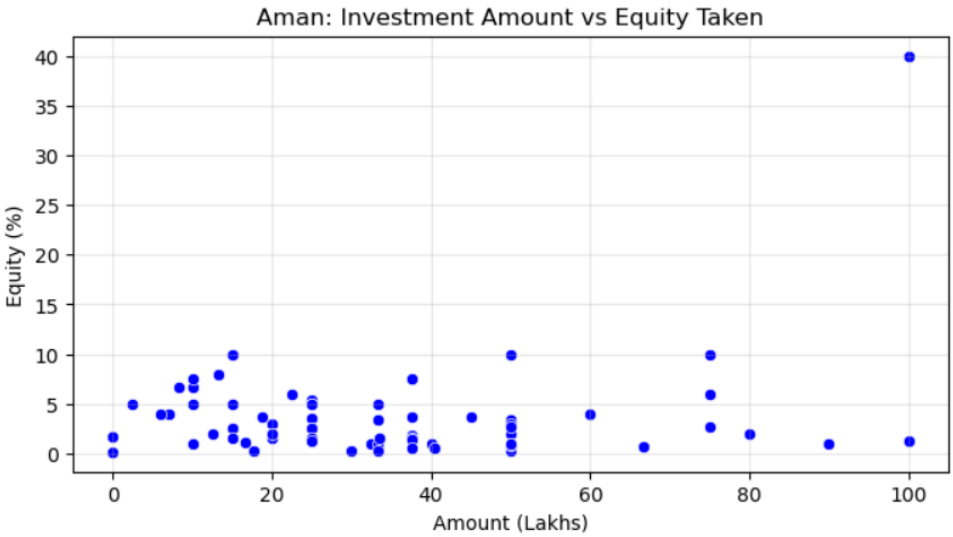
```
# 1. Your list of 7 sharks
sharks = ['Ghazal', 'Vineeta', 'Ashneer', 'Amit', 'Aman', 'Peyush', 'Namita']

# 2. A simple loop to create 7 separate plots
for shark in sharks:
    # Create a new blank chart area for each shark
    plt.figure(figsize=(8, 4))

    # Draw the dots
    sns.scatterplot(
        data=df,
        x=f'{shark} Investment Amount',
        y=f'{shark} Investment Equity',
        color='blue'
    )

    # Add labels so it looks professional
    plt.title(f'{shark}: Investment Amount vs Equity Taken')
    plt.xlabel('Amount (Lakhs)')
    plt.ylabel('Equity (%)')
    plt.grid(True, alpha=0.3)

    # Show the chart and then clear it for the next shark
    plt.show()
```

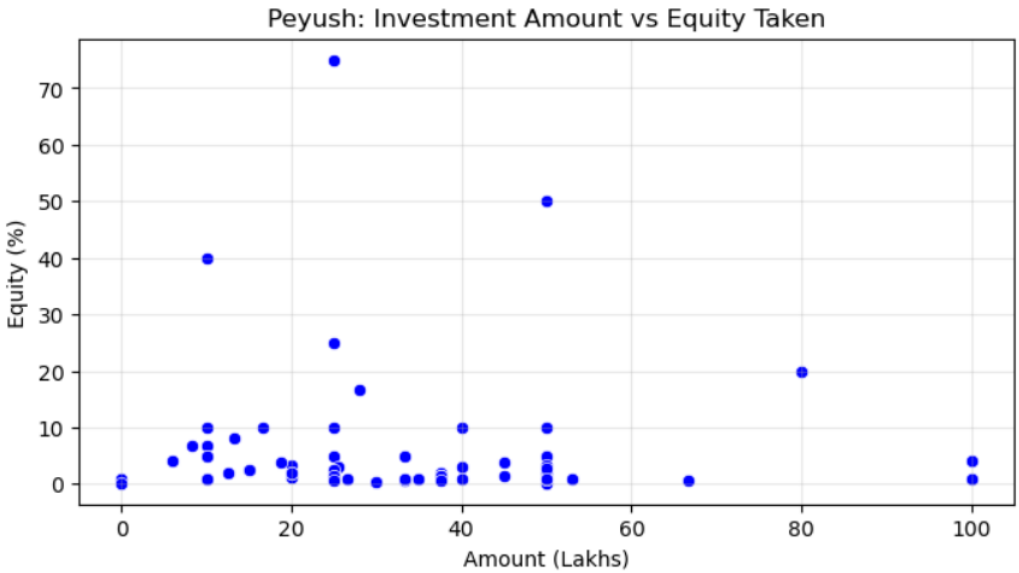

Ghazal: Investment Amount vs Equity Taken
Equity (%)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0
5
10
15
20
25
30
Amount (Lakhs)

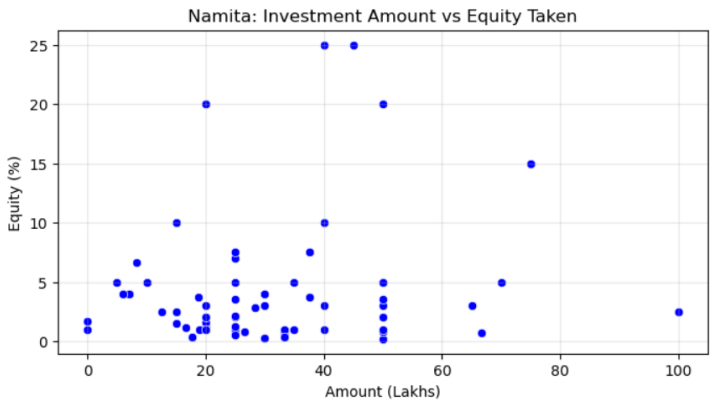

Vineeta: Investment Amount vs Equity Taken
Equity (%)
25
20
15
10
5
0
20
40
60
80
100
Amount (Lakhs)


Ashneer: Investment Amount vs Equity Taken
Equity (%)
25
20
15
10
5
0
0
10
20
30
40
50
60
70
Amount (Lakhs)


Amit: Investment Amount vs Equity Taken
Equity (%)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
20
40
60
80
100
Amount (Lakhs)


Aman: Investment Amount vs Equity Taken
Equity (%)
40
35
30
25
20
15
10
5
0
0
20
40
60
80
100
Amount (Lakhs)


Peyush: Investment Amount vs Equity Taken
Equity (%)
70
60
50
40
30
20
10
0
0
20
40
60
80
100
Amount (Lakhs)


Namita: Investment Amount vs Equity Taken
Equity (%)
Amount (Lakhs)
0
5
10
15
20
25
0
20
40
60
80
100

In [ ]: